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ROURKELA

PROJECT DOCUMENTATION

CUPGS Academic Feedback Manager

Anonymous Student Feedback System
with Role-Based Access Control

INSTITUTION

Centre for Urban & Planning
Graduate Studies (CUPGS)
BPUT Rourkela, Odisha

SYSTEM TYPE

Full-Stack Web Application
Progressive Web App (PWA)
Cross-Platform · Dark Theme

TECHNOLOGY STACK

React 18 · TypeScript · Node.js · Express · PostgreSQL · Drizzle ORM · Tailwind CSS · jsPDF

Academic Year 2024-25 · Version 1.0 · Confidential

DEPARTMENTS COVERED

- Computer Science & Engineering (CSE)
- Electronics & Communication Engineering (ECE)
- Electrical Engineering (EE)
- Mechanical Engineering (ME)
- Civil Engineering (CE)
- Civil Engineering (CE)

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1. Project Overview & Objectives

What is CUPGS Feedback Manager?

CUPGS Feedback Manager ek full-stack web application hai jo BPUT Rourkela ke CUPGS (Centre for Urban & Planning Graduate Studies) ke liye banaya gaya hai. Yeh system students ko apne courses aur faculty ke baare mein anonymous feedback dene ki suvidha deta hai, aur Heads of Departments (HODs) ko real-time analytics aur PDF reports generate karne ki suvidha deta hai.

KYA HAI (What)

- Anonymous feedback system
- Drag-based half-star rating (0.5 steps)
- HOD-customizable question forms
- Real-time analytics dashboard
- PDF report generation
- Role-based access control

KAUN USE KARTA HAI (Who)

- Students — feedback submit karo
- Faculty — apna performance dekho
- HOD — analytics + form customize
- Admin — pura system manage karo

KYUN BANAYA (Why)

- Manual paper feedback se hata
- Anonymous hone se honest feedback
- HOD ko real data milta hai
- Faculty improvement track hota hai
- Transparency badti hai

1.1. Core Objectives

Anonymous Feedback
Students apna feedback bina naam bataye de sakte hain. Reference ID se track kar sakte hain lekin faculty/HOD identify nahi kar sakta.

Half-Star Ratings
Drag-based rating system 0.5 increment mein — 0.5 se 5.0 tak. Teaching Quality, Course Content, Lab Facilities, Study Material, Overall.

HOD Analytics Dashboard
HOD apne department ke sab courses ki ratings dekh sakta hai, trends compare kar sakta hai, aur professional PDF report download kar sakta hai.

Customizable Feedback Forms
HOD apne department ke liye custom questions add kar sakta hai — text, rating, ya multiple choice. Yeh standard questions ke saath milke show hote hain.

Faculty-Course Integrity
System ensure karta hai ki feedback sirf us course ke assigned faculty ke liye diya jaye. Cross-department manipulation prevent hota hai.

Feedback Window Control
Admin specific time periods ke liye feedback windows khol aur band kar sakta hai. Window band hone pe feedback submit nahi hoga.

1.2. System Scope

IN SCOPE:

- 5 Engineering departments — CSE, ECE, EE, ME, CE
- Odd semester (July–Oct) aur Even semester (Nov–June) feedback
- Academic year auto-calculation (April = new year start)
- Complete audit trail with reference IDs

OUT OF SCOPE:

- Online examination system
- Attendance management
- Fee payment portal
- Result management system

1.3. Key Performance Indicators

268+	5	4	5+	100%	0.5
Courses	Departments	User Roles	Rating Categories	Anonymous	Rating Steps

1.4. Academic Context

Semester System:

ODD Semester (July – October)
1st, 3rd, 5th, 7th semester courses — feedback window opens October

EVEN Semester (November – June)
2nd, 4th, 6th, 8th semester courses — feedback window opens June

Academic Year Calculation:

If current month >= April (month ≥ 4) Academic Year = currentYear – (currentYear+1) e.g., "2025-26"
If current month < April (Jan-Mar) Academic Year = (currentYear-1) – currentYear e.g., "2024-25"

Login Credentials Format:

Student	ID: **CUPGS/DEPT/NNN (e.g. CUPGS/CSE/042)**	PIN/Pass: **1234 (PIN)**
Faculty	ID: **CUPGS/DEPT/NNN (e.g. CUPGS/CSE/101)**	PIN/Pass: **1234 (PIN)**
HOD	ID: **HOD/DEPT/001 (e.g. HOD/CSE/001)**	PIN/Pass: **DEPT@2025 (e.g. CSE@2025)**
Admin	ID: **bput@admin2025**	PIN/Pass: **Admin password**

2. Problem Statement & Solution Approach

2.1. Problem Statement

Existing Problems at CUPGS:

- Manual paper-based feedback system — time-consuming, error-prone, hard to aggregate
- No anonymity guarantee — students identify hone ke darr se honest feedback nahi dete
- HOD ko manually sab forms collect aur calculate karne padte hain — weeks lag jaate hain
- Faculty-wise ya course-wise comparison possible nahi tha — koi analytics nahi
- Custom questions add karna impossible tha — ek hi standard form sabke liye

2.2. Solution — How CUPGS Feedback Manager Solves It

[NO] Paper forms - no digitization	[OK] 100% digital — web-based form submit, PostgreSQL mein store, instant available
[NO] No anonymity — fear of identification	[OK] Reference ID system — sirf student jaanta hai apna code, faculty/HOD identify nahi kar sakta
[NO] Manual aggregation — weeks of work	[OK] Real-time HOD dashboard — ek click mein department analytics, PDF report download
[NO] No course-wise comparison	[OK] Per-course rating breakdown — 5 categories mein har course ki average rating visible
[NO] Standard form only	[OK] HOD form builder — custom text/rating/choice questions add kar sakta hai apne dept ke liye
[NO] No access control on timing	[OK] Feedback window system — Admin specific dates pe window khol/band kar sakta hai

2.3. Unique Features Not Found in Typical Systems

Half-Star Drag Rating
Sirf click nahi — drag karke 0.5 step pe rating set karo. Mouse aur touch dono pe kaam karta hai. Visually engaging aur precise.

Live HOD Analytics
HOD dashboard real-time data show karta hai — department average, per-course breakdown, rating category comparison, top/bottom performers.

Faculty-Course Integrity
Agar course mein faculty assign nahi hai (Faculty TBA), toh feedback dene pe faculty section optional rehta hai. Assigned courses mein auto-lock hota hai.

Reference ID Anonymity
Har submission pe unique FB-YYYYMMDD-XXXXXX format ka reference ID milta hai. Student apna feedback verify kar sakta hai, koi aur nahi.

HOD Form Builder
HOD apne dashboard se custom questions add/remove kar sakta hai bina developer ke. Fields: short text, long text, single choice, rating scale.

One-Click PDF Report
HOD 'Download PDF' click kare toh complete department report milti hai — college header, all courses, ratings table, charts, semester info.

2.4. Comparison with Traditional System

Feature	Traditional Paper System	CUPGS Feedback Manager
Feedback Submission	[N] Physical form + pen	[Y] Digital web form, any device
Anonymity	[N] None — handwriting visible	[Y] 100% — reference ID only
Result Time	[N] 2–4 weeks manual count	[Y] Instant — real-time dashboard
Custom Questions	[N] Reprint forms needed	[Y] HOD adds in 1 click
Rating System	[N] 1–5 whole numbers only	[Y] 0.5–5.0 half-star drag
Report Format	[N] Handwritten summary	[Y] Auto-generated PDF report
Access Control	[N] No timing control	[Y] Admin-controlled windows
Data Storage	[N] Physical files (losable)	[Y] PostgreSQL — permanent, secure

3. System Architecture & Features

3.1. Three-Tier Architecture

TIER 1 — PRESENTATION LAYER (Frontend)
React 18 + TypeScript + Vite + Tailwind CSS + shadcn/ui components
Pages: Login · Submit Feedback · HOD Dashboard · Admin Panel · Faculty View
PWA: Installable, Offline-capable, Responsive (Mobile + Desktop + Tablet)

[REST API -- HTTP/JSON -- Axios]

TIER 2 — APPLICATION LAYER (Backend API)
Node.js + Express.js + TypeScript · Runs on PORT 8080
Routes: /api/auth · /api/feedback · /api/departments · /api/courses · /api/faculty · /api/windows
Middleware: Rate Limiting · Session Management · Input Validation (Zod) · Trust Proxy

[Drizzle ORM -- SQL Queries]

TIER 3 — DATA LAYER (Database)
PostgreSQL · 6 Tables · Replit managed database
Tables: departments · courses · faculty · feedback · form_templates · feedback_windows
Features: Real FK constraints · JSONB custom answers · real type for 0.5 ratings

3.2. Frontend Pages & Routing

/	**Login Page** — Role-select karo (Student/Faculty/HOD/Admin) ID aur PIN enter karo dashboard pe redirect
/submit-feedback	**Submit Feedback** — Course select karo semester auto-fill drag-star rating comments submit Ref ID milega
/hod-dashboard	**HOD Dashboard** — Department analytics, per-course rating table, PDF download, feedback window toggle, form builder
/admin	**Admin Panel** — Institution-wide analytics, all departments overview, course/faculty management, window control
/faculty	**Faculty View** — Apne assigned courses ka feedback dekho, rating breakdown, student comments (anonymous)

3.3. API Endpoints Overview

Method	Endpoint	Purpose
POST	`/api/auth/login`	Student/Faculty/HOD/Admin login
POST	`/api/feedback`	Submit anonymous feedback
GET	`/api/feedback/check/:refId`	Reference ID se apna feedback verify karo
GET	`/api/departments/:id/hod-report`	HOD analytics data fetch karo
GET	`/api/departments/:id/form-template`	Custom form template load karo
POST	`/api/departments/:id/form-template`	HOD form save karo
GET	`/api/windows`	Active feedback windows list
POST	`/api/windows`	Admin — new window create karo
GET	`/api/analytics/dashboard`	Admin institution-wide analytics
GET	`/api/courses`	Courses list with faculty info

3.4. Rate Limiting & Security

Global Rate Limit	100 requests per 15 minutes per IP — DoS attack se protection
Auth Rate Limit	10 login attempts per 15 minutes — brute force attack prevent
Feedback Rate Limit	20 submissions per 10 minutes — spam prevent
Session Security	Express-session + SESSION_SECRET environment variable mein store
[OK] Input Validation	Zod schema validation — har route pe type-safe input check
Trust Proxy	app.set('trust proxy', 1) — reverse proxy ke peeche correct IP

4.1. Role Overview

STUDENT	FACULTY	HOD	ADMIN

STUDENT ROLE

Login: CUPGS/DEPT/NNN | PIN: 4-digit PIN (default: 1234)

[OK] CAN DO:
- Submit anonymous feedback for enrolled courses
- Drag-based half-star rating (0.5 to 5.0)
- Add text comments per category
- Answer HOD's custom questions
- Get unique Reference ID after submit

[NO] CANNOT DO:
- See other students' feedback
- See faculty names in analytics
- Access HOD or Admin panels
- Submit feedback outside active windows

FACULTY ROLE

Login: CUPGS/DEPT/NNN (same format) | PIN: 4-digit PIN

[OK] CAN DO:
- View own course feedback analytics
- See average ratings per category
- See anonymous student comments
- Compare own performance across semesters
- View rating distribution charts

[NO] CANNOT DO:
- Submit feedback
- See which student gave which rating
- Access other faculty's feedback
- Access HOD/Admin features

HOD ROLE

Login: HOD/DEPT/001 | PIN: DEPT@2025 (e.g., CSE@2025)

[OK] CAN DO:
- View entire department analytics dashboard
- See per-course rating breakdown (5 categories)
- Compare all courses in department
- Download PDF report (one click)
- Customize feedback form (add/remove questions)

[NO] CANNOT DO:
- See individual student responses
- Access other departments' data
- Create/delete courses or faculty
- Modify Admin-level settings

ADMIN ROLE

Login: bput@admin2025 | PIN: Admin password

[OK] CAN DO:
- Institution-wide analytics dashboard
- All departments combined overview
- Create/edit/delete departments
- Manage courses (add, assign faculty)
- Manage faculty records

[NO] CANNOT DO:
- See individual anonymous feedback content
- Override HOD's custom form for a dept

5.1. Authentication Module

Authentication Module

Role-based login system jo different dashboard pe redirect karta hai.

Step-by-Step Flow:

1. User apna Role select karta hai (Student / Faculty / HOD / Admin)
2. ID aur PIN/Password enter karta hai
3. Server ID pattern check karta hai — CUPGS/DEPT/NNN ya HOD/DEPT/001
4. Valid hone pe session create hota hai, invalid pe error message
5. Session-based auth — browser close karne pe auto logout
6. Rate limit: 10 failed attempts per 15 min per IP ke baad block

5.2. Feedback Submission Module

Feedback Submission Module

Core module — student ka anonymous feedback collect karta hai.

Step-by-Step Flow:

1. Student apna Department aur Course select karta hai
2. Faculty automatically course se assign hoti hai (ya TBA show hota)
3. Semester aur Academic Year auto-calculate hota hai
4. Drag-based star rating — 5 categories mein 0.5 step pe
5. Text comments optional — each category ke liye
6. HOD ke custom questions (agar configured ho) appear hote hain
7. Submit pe: server validate integrity check store Ref ID return

5.3. HOD Analytics & Dashboard Module

HOD Analytics & Dashboard Module

HOD ka main tool — real-time department performance analytics.

Step-by-Step Flow:

1. Department summary: total feedback count, overall average rating
2. Per-course table: har course ki 5-category rating breakdown
3. Sorting: best/worst performing courses identify karo
4. Trend view: semester-wise comparison
5. PDF Report: college header, date, all courses, ratings, download
6. Form Builder: custom questions add/remove/reorder karo
7. Window status: active hai ya nahi, dates dikhai deti hain

5.4. Admin Institution Management Module

Admin Institution Management Module

Admin ka complete control — institution-wide management.

Step-by-Step Flow:

1. Department-wise analytics summary (all 5 departments)
2. Highest/lowest rated courses across institution
3. Course management: add, edit, assign faculty, deactivate
4. Faculty management: add, edit, department assign
5. Feedback Window: create with title, dates, dept (or global)
6. Window control: toggle active/inactive, delete windows
7. Cross-department comparison analytics

6.1. Complete Technology Stack

FRONTEND STACK

React 18	UI library — component-based, hooks-based state management
TypeScript	Type-safe JavaScript — compile-time error catching
Vite	Build tool — lightning-fast HMR, optimized production builds
Tailwind CSS	Utility-first CSS — responsive, dark theme, custom design
shadcn/ui	Accessible component library — buttons, cards, dialogs, forms
React Query	Server state management — caching, background updates, loading states
React Hook Form	Form management — validation, error handling, submit state
sPDF + html2canva	PDF generation — HOD report, one-click download
Framer Motion	Animations — smooth transitions, drag interactions

BACKEND STACK

Node.js	JavaScript runtime — non-blocking I/O, event-driven
Express.js	HTTP server framework — routing, middleware, sessions
TypeScript	Type safety on server too — shared types with frontend
express-session	Server-side session management — authentication state
express-rate-limit	API rate limiting — brute force + DoS protection
Zod	Schema validation library — parse and validate all API inputs
Pino	Structured logging — JSON format, production-grade performance

DATABASE STACK

PostgreSQL	Relational database — ACID compliant, JSON support, production-grade
Drizzle ORM	Type-safe ORM — schema-first, SQL-like queries, no magic
Drizzle Kit	Database migrations — push schema changes safely
JSONB columns	feedback.custom_answers aur form_templates.fields ke liye
real type	Rating columns 0.5 steps support karne ke liye float precision

INFRASTRUCTURE STACK

Replit	Cloud development + hosting platform
pnpm workspaces	Monorepo — api-server aur bput-feedback packages
PWA (Manifest+SW)	Installable app, offline capability, mobile-optimized
Environment secrets	SESSION_SECRET securely stored, never in code

6.2. UI Design System

Dark Glassmorphism Theme

Background: #0f172a (Slate 900) · Cards: backdrop-blur + rgba white overlay

Primary: #1e3a8a Blue · Accent: #2563eb · Success: #14532d Green · Error: #dc2626 Red

7. How To Use The System (Step-by-Step)

STUDENT — Step by Step Guide

1. **Open App**
   Browser mein app URL open karo ya installed PWA launch karo
2. **Select Role**
   'Student' select karo login page pe
3. **Login**
   ID: CUPGS/CSE/042 (example), PIN: 1234 Login karo
4. **Select Course**
   Department Course dropdown se apna course choose karo
5. **Rate Faculty**
   Har category ke liye star rating drag karo (0.5 step mein)
6. **Add Comments**
   Optional text comments add karo (har category ke liye)
7. **Custom Questions**
   Agar HOD ne custom questions add kiye hain, unhe bhi fill karo
8. **Submit**
   'Submit Feedback' button dabao Reference ID note karo
9. **Verify (Optional)**
   Reference ID se apna submission verify kar sakte ho

HOD — Step by Step Guide

1. **Login**
   ID: HOD/CSE/001, Password: CSE@2025
2. **Dashboard View**
   Department analytics automatically load hogi
3. **View Analytics**
   Per-course rating table dekho — 5 categories mein breakdown
4. **Filter/Sort**
   Courses ko sort karo — best/worst performing
5. **Download PDF**
   'Download PDF Report' click karo — complete report milegi
6. **Form Builder**
   'Customize Form' section mein custom questions add karo
7. **Window Status**
   Active feedback window hai ya nahi — dates dekho

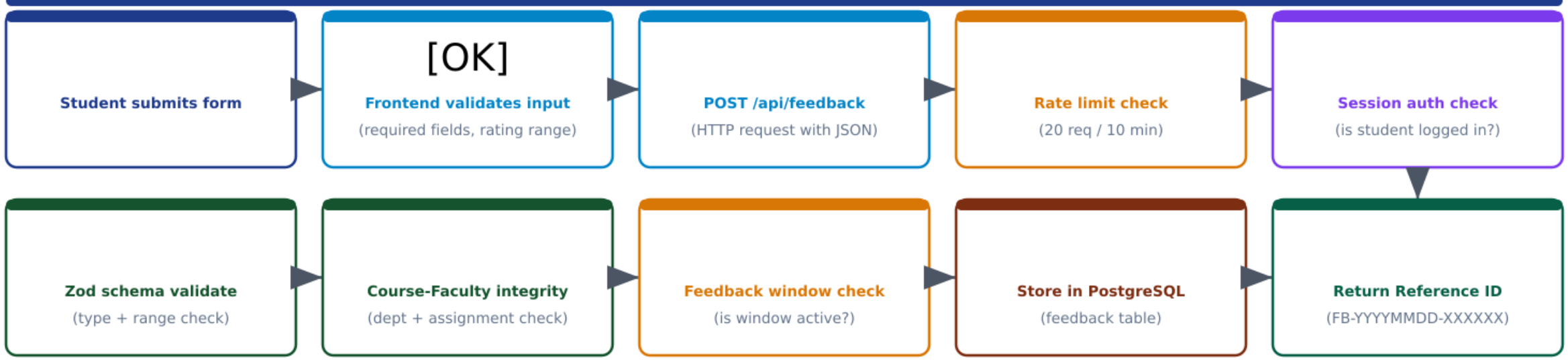
ADMIN — Step by Step Guide

1. **Login**
   ID: bput@admin2025, Password enter karo
2. **Institution Dashboard**
   All departments ka combined analytics dekho
3. **Manage Courses**
   Course add karo, faculty assign karo, semester set karo
4. **Manage Faculty**
   Faculty records add/edit/delete karo
5. **Feedback Windows**
   New window create karo — title, start date, end date set karo
6. **Toggle Window**
   Window active/inactive karo — students control ho
7. **Cross-Dept View**
   Sab departments ka performance comparison dekho

FACULTY — Step by Step Guide

1. **Login**
   ID: CUPGS/CSE/101 (example), PIN: 1234
2. **My Courses**
   Apne assigned courses ki list dekho
3. **View Ratings**
   Har course ki average rating aur breakdown dekho
4. **Read Comments**
   Anonymous student comments padhte hain (no names shown)
5. **Track Progress**
   Semester-wise performance compare karo

8.1. Feedback Submission Data Flow



8.2. Database Schema — Complete

departments

- id PK
- code UNIQUE
- name
- hod_name
- hod_employee_id
- created_at

faculty

- id PK
- department_id FK
- employee_id UNIQUE
- name
- designation
- email
- is_active

courses

- id PK
- department_id FK
- faculty_id FK?
- code UNIQUE
- name
- semester
- academic_year
- credits

feedback

- id PK
- reference_id UNIQUE
- course_id FK
- faculty_id FK?
- department_id FK
- semester
- 5× real ratings
- comments

form_templates

- id PK
- department_id FK UNIQUE
- fields JSONB
- created_at
- updated_at

feedback_windows

- id PK
- department_id FK?
- title
- is_active
- start_date
- end_date

8.3. Security Measures

Session Management	Server-side express-session — SESSION_SECRET environment variable mein, never in code. Browser mein sirf session cookie store hoti hai.
Rate Limiting (3 levels)	Global: 100/15min · Auth: 10/15min · Feedback POST: 20/10min. Trust proxy enabled for correct IP behind reverse proxy.
[OK] Input Validation	Zod schemas har API route pe — types, ranges, required fields. Invalid input pe clear error message, server crash nahi hota.
Anonymous Guarantee	Server kabhi student ID aur feedback ID link karke store nahi karta. Reference ID cryptographically unique — guessing se track nahi hota.
Faculty-Course Integrity	Cross-department feedback manipulation impossible — server course ka department aur student ka department match karta hai.
Window Enforcement	Server-side window check — client-side bypass possible nahi. Server database se active window check karta hai har submission pe.
No Direct DB Access	Database sirf Drizzle ORM ke through access hoti hai. Raw SQL queries nahi — SQL injection practically impossible.
CORS + Proxy Config	Vite dev proxy configure hai — frontend-backend same origin pe, CORS issues nahi. Production mein Express sirf trusted paths serve karta hai.

8.4. Anonymity Architecture

How Anonymity is Guaranteed:
Jab student feedback submit karta hai, server ek cryptographically random Reference ID generate karta hai (format: FB-YYYYMMDD-XXXXXX). Yeh ID feedback record mein store hoti hai, lekin student ka login ID (CUPGS/CSE/042) kabhi feedback table mein store nahi hota.

What is STORED in feedback table:
reference_id, course_id, department_id, semester, academic_year, student_year, section, ratings, comments, custom_answers

What is NEVER STORED in feedback table:
student_id, employee_id, name, login credentials, IP address, browser info — NOTHING that identifies the student

[OK] Even the developer / DBA cannot trace feedback back to a specific student.
[OK] Student can verify OWN feedback using their Reference ID — but others cannot use it to identify them.

8.5. Error Handling

9.1. Project Summary

CUPGS Academic Feedback Manager ek complete full-stack web application hai jo BPUT Rourkela ke CUPGS department ke liye specially design kiya gaya hai. Iska main purpose hai ki students apne courses aur faculty ke baare mein honest, anonymous feedback de sakein — bina kisi fear ke — aur HOD is data ko real-time analytics aur professional PDF reports mein dekh sakein.

System mein 4 roles hain — Student, Faculty, HOD, Admin. Har role ke liye alag dashboard aur permissions hain. Data PostgreSQL mein securely store hota hai, aur anonymity guarantee ki gayi hai reference ID system se.

9.2. What Was Achieved (Key Deliverables)

[OK] Anonymous Feedback System
Students kisi bhi device se feedback de sakte hain, 100% anonymously

[OK] Half-Star Drag Rating
5 categories mein 0.5-step precise ratings — better data quality

[OK] Real-time HOD Dashboard
Department analytics, per-course comparison, instant visibility

[OK] One-click PDF Reports
HOD ko manual work se mukti — complete PDF ek click mein

[OK] HOD Form Builder
Custom questions bina developer ke — HOD apne hisaab se form customize kar sakta hai

[OK] Faculty-Course Integrity
Cross-department fraud prevent — system automatically validate karta hai

[OK] Admin Institution Control
Centralized management — departments, courses, faculty, windows sab ek jagah

[OK] Feedback Window System
Time-controlled feedback collection — Admin ke haath mein complete control

[OK] Progressive Web App (PWA)
Installable, responsive, mobile-friendly — koi app store nahi chahiye

[OK] Rate Limiting + Security
Production-grade security — brute force, DoS, SQL injection protection

9.3. Future Scope & Enhancements

Phase 2 — Short Term

- Email notifications — HOD ko automatically report milegi semester end pe
- Graphical charts in PDF — bar charts, radar charts for visual comparison
- Student feedback history — apne purane submissions dekho (via Ref IDs)
- Multi-language support — Odia, Hindi mein bhi available ho
- Bulk course import — Excel se courses directly import karo

Phase 3 — Medium Term

- AI-powered sentiment analysis — comments pe automatic positive/negative tagging
- Trend analysis — semester-over-semester performance graphs for faculty
- Mobile app (React Native) — iOS aur Android native app
- Comparison benchmarking — national/state level comparison with similar institutions
- Automated action triggers — low rating pe HOD ko auto-alert system

Phase 4 — Long Term

- Institution network — multiple colleges ek platform pe
- NAAC/NIRF integration — feedback data directly NAAC reports ke liye exportable
- Predictive analytics — ML model se at-risk courses early identify karo
- Video feedback — students short video testimonials de sakein (anonymous)
- Blockchain anonymity — on-chain proof of anonymity for ultimate trust

9.4. Technical Lessons & Best Practices Used

- Monorepo structure (pnpm workspaces) — frontend aur backend ek sath manage
- Type safety end-to-end — TypeScript + Zod + Drizzle = zero runtime type errors in theory
- Separation of concerns — API routes, business logic, database layer alag-alag
- Security by default — rate limiting, session management, input validation from day 1
- DRY principle — shared types between frontend and backend
- Git-based version control — every feature is a checkpoint